

Assignment: MCQ Questions based on Cars24 Case Study

Answer Key with Explanations

Q1. A - A. `df.head(5)` returns the first 5 rows. This is used to preview the top of the DataFrame.

Q2. C - C. `df.info()` gives column names, data types, non-null counts ? useful for overall structure.

Q3. A - A. `df.describe(include='object')` gives statistics like count, unique, top, and freq for categorical data.

Q4. B - B. `df.tail(3)` returns the last 3 rows of the DataFrame.

Q5. C - C. `df.describe()` does not give median; it gives count, mean, std, min, 25%, 50%, 75%, and max.

Q6. C - C. `.plot.density()` creates a KDE (Kernel Density Estimation) plot to show distribution.

Q7. C - C. Box plots are best for detecting outliers and spread of data using IQR.

Q8. D - D. Whiskers represent minimum and maximum values excluding outliers in a boxplot.

Q9. B - B. This setup is used to study how categories affect numeric values (like `selling_price`).

Q10. A - A. Heatmaps with missing value indicators can help visualize the presence of nulls.

Q11. B - B. To replace 0s with NaNs, use `replace(0, np.nan)`.

Q12. C - C. `df.drop('col', axis=1)` permanently removes the column.

Q13. B - B. Median is resistant to outliers, unlike mean.

Q14. C - C. If only 0.8% have `mileage = 0`, we can drop or impute safely.

Q15. D - D. `apply(lambda x: ...)` lets us split strings into custom columns.

Q16. A - A. `age = 2020 - df['year']` gives the car's age from manufacture year.

Q17. C - C. `sample(frac=1)` shuffles the entire DataFrame.

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- Q18. B - B. Log transformation is commonly used to reduce right-skewed data.
- Q19. C - C. Heatmap is a visual encoding technique, not an encoding algorithm.
- Q20. B - B. One-hot encoding increases dimensionality by creating new columns per category.
- Q21. C - C. Target encoding replaces categories with mean of target variable.
- Q22. A - A. ``get_dummies`` is used to create one-hot encodings, often with a prefix.
- Q23. C - C. Heatmaps show matrix-based correlations using color gradients.
- Q24. D - D. Both ``.corr()`` and ``df['x'].corr(df['y'])`` work.
- Q25. C - C. Seaborn is typically aliased as ``sns`` and used for statistical visualization.
- Q26. B - B. ``plt.xticks(rotation=90)`` rotates labels to prevent overlap.
- Q27. C - C. The ``model`` is extracted from everything **except** the first word (brand).
- Q28. A - A. ``drop()`` errors out if the column is missing unless ``errors='ignore'`` is used.
- Q29. B - B. ``pd.read()`` doesn't exist. It should be ``pd.read_csv()``.
- Q30. B - B. ``df.columns()`` is invalid since ``columns`` is a property, not a method.
- Q1. B - B. ``normalize=True`` gives proportion (percentages), not raw counts.
- Q2. B - B. Pairplot shows pairwise distribution of features with plots.
- Q3. C - C. ``concat(axis=1)`` combines datasets side by side.
- Q4. C - C. ``df.shape`` returns a tuple of (rows, columns).
- Q5. B - B. Drop rows with many missing values to reduce noise.
- Q6. A - A. ``hist().plot()`` is improper chaining, though might work in some contexts. Prefer ``plt.show()``.

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- Q7. C - C. Using `.map()` to assign numeric values from dictionary is label encoding.
- Q8. D - D. All three options correctly remove the column.
- Q9. C - C. Target encoding works better with many categories since it doesn't expand column space.
- Q10. C - C. Boxplot shows the spread and outliers for each brand vs price.
- Q11. B - B. `.density()` is not a valid method on the histogram result.
- Q12. B - B. Mapping groupby mean of `selling_price` creates price-informed encoding.
- Q13. B - B. `df.sort()` doesn't exist ? should use `sort_values()`.
- Q14. B - B. `iloc[-1]` accesses the last row of the DataFrame.
- Q15. A - A. `hue` adds multiple KDE lines based on the category.
- Q16. D - D. Dropping one dummy avoids the dummy variable trap (multicollinearity).
- Q17. A - A. Log transformation compresses high values, reducing skewness.
- Q18. B - B. `ascending=False` sorts descending.
- Q19. B - B. `value_counts()` shows frequency of each unique value.
- Q20. A - A. Boxplot helps compare numerical stats grouped by category like brand.